

A polymerization closure for the membrane Brownian ratchet

Per-tip, gap- & force-gated monomer addition — replacing the ad-hoc `stericPolyFactor`

BoxOfActin · design note · 2026-06-15 · context: lamellipodium P1–P3; Mogilner–Oster elastic Brownian ratchet

1. Why a closure, not emergence

The elastic ratchet's tip fluctuation lives on the monomer scale ($\delta \approx 2.7$ nm). Our filaments are chains of rigid `FilSegment` rods, each spanning `stdSegLength` ≈ 32 –64 monomers, so the terminal segment is rigid over ~ 90 –170 nm. The tip therefore cannot fluctuate at the scale the ratchet needs, and joint bending only opens a membrane-normal gap for *oblique* filaments (the normal excursion per joint rotation scales as $L \sin \theta d\theta$, vanishing at normal incidence). The mechanism is genuinely **sub-grid**. The disciplined response is a physically-parameterized closure — not a "flexible tip" hack, which would reintroduce free parameters, miss the density-dependent free length, destabilize the integrator, and still fail head-on filaments.

A closure does **not** kill emergence. Supplied as a per-tip law that depends on the tip's actual geometry and load, it leaves the interesting behavior — $\pm 35^\circ$ orientation selection, load-sharing, array treadmilling — to self-organize at the *network* level (which filaments grow vs. stall vs. cap).

2. The core relation — the penalty is on the deficit

The rate is gated by whether there is room for a monomer of length δ . If the barbed end already sits a gap g short of the membrane (measured along the addition direction), a thermal fluctuation only needs to open the **remaining** ($\delta - g$), not the whole δ :

$$k_{\text{poly}} = k_{\text{on}} c \cdot \begin{cases} 1, & g \geq \delta \quad (\text{room already exists — free rate}) \\ \exp\left(-\frac{f(\delta - g)}{k_B T}\right), & 0 \leq g < \delta \quad (\text{ratchet against load } f) \end{cases}$$

Limit checks:

- $g = 0$: $k_{\text{poly}} = k_{\text{on}} c \exp(-f\delta/k_B T)$ — recovers classic Mogilner–Oster.
- $g = \delta$: $k_{\text{poly}} = k_{\text{on}} c$ — fully unobstructed; the penalty turns *off*.
- monotonic and continuous in g between — an existing gap raises the rate, exactly as expected.

Depolymerization is left load-independent (the load doesn't help remove a bound monomer), so the full force–velocity follows as

$$v = \delta (k_{\text{poly}} - k_{\text{off}}), \quad f_{\text{stall}} \delta = k_B T \ln \frac{c k_{\text{on}}}{k_{\text{off}}},$$

i.e. the load can do work per monomer up to the polymerization free energy, then stalls — the energy is the chemical drive you already supply, not a new input.

3. Incidence angle enters through the geometry of "room"

The monomer adds along the filament axis $\hat{u}$; what matters is the **axial** room before the tip meets the membrane. With θ the angle from the membrane normal $\hat{n}$,

$$g_{\text{axial}} = \frac{g_{\perp}}{\cos \theta},$$

so an oblique tip converts a given perpendicular gap into *more* axial room (a grazing tip slides along the wall rather than into it), while the compliance available to close the residual deficit is the angle-projected bending mode $\sim \sin \theta$. Thus θ enters twice — how much room a gap buys ($\cos \theta$) and how easily the deficit is closed ($\sin \theta$) — and both are computable per tip.

4. The payoff: geometry does most of the work

The gap g is essentially `end2TipC`, the tip clearance the model already computes. Critically, **gap and contact force are anti-correlated**: when a real gap exists the tip isn't touching the membrane, so $f \approx 0$ and the rate is near-free *without* the closure doing anything. The analytic Boltzmann factor only matters in the thin regime where $g < \delta$ and the tip is loaded — the last sub-monomer crunch.

This shrinks the closure to a small correction on top of geometry the grid resolves natively. The noisy per-tip **force** read — the main implementation risk — only enters in that sub- δ sliver; everywhere else `end2TipC` alone gives the answer.

5. One honest subtlety — what `end2TipC` represents

Whether `end2TipC` is the *instantaneous* or the *quasi-static mean* gap depends on whether the tip's fast fluctuation is resolved. With the over-damped rigid segment it is effectively the **mean** gap (set by the balance of polymerization push and membrane load). So the clean reading is: use `end2TipC` as the resolved mean clearance, let geometry handle $g \geq \delta$ for free, and let $\exp(-f(\delta - g)/k_B T)$ supply precisely the **sub-grid thermal wobble** the rigid segment cannot generate, over the residual $(\delta - g)$. Geometry where the grid resolves it; analytic closure only for the unresolved fluctuation.

6. Inputs (all available per tip)

Symbol	Meaning	Source in the model
g	clearance to membrane along the addition direction	<code>end2TipC</code> (registered in the membrane collision phase), divided by $\cos \theta$ for the axial value
f	membrane-normal load on the tip	reaction force from the membrane point/face collision (the noisy read — only needed when $g < \delta$)
θ	incidence angle from membrane normal	$\hat{\mathbf{u}} \cdot \hat{\mathbf{n}}$; $\hat{\mathbf{n}}$ from the local membrane (linked-neighbour PCA normal, as the viewer already derives)
L_{free}	free length to nearest clamp (sets compliance)	tip-to-nearest- <code>Arp2/3</code> distance — inherits the local branch density automatically
δ	monomer increment	<code>halfmono</code> (the existing poly step)

7. Where it slots in

Replace the `stericPolyFactor` multiplier in `FilSegment.end2BiochemSim` with this gate: compute g_{axial} from `end2TipC` and $\cos \theta$; if $g_{\text{axial}} \geq \delta$, no attenuation; else multiply the poly rate by $\exp(-f(\delta - g_{\text{axial}})/k_B T)$. Crucially, the membrane *collision* must keep providing mechanical push-back but stop separately throttling polymerization — otherwise the load is double-counted. Default-off behind a flag so it can be A/B'd against the current soft proxy.

Net. A per-tip law gated on the real clearance, with the ratchet penalty applied only to the unresolved sub-monomer deficit and weighted by the local load and incidence angle. More faithful than `stericPolyFactor`, discretization-independent, and *less* reliant on the messy force read than a blanket $\exp(-f\delta/k_B T)$ would be — because an existing gap, at whatever angle, does exactly what you said it must: it raises the polymerization probability, all the way to the free rate once it reaches a full monomer.